Telecom Customer Churn Analysis - Project Insights & Skills Implemented

Project Overview

Built an end-to-end Telecom Customer Churn Analysis project using Python, MySQL, and Power BI to understand customer behavior, identify churn patterns, and provide business recommendations for customer retention.

Dataset Summary

The project used the Telco Customer Churn dataset containing customer demographics, services subscribed, contract information, payment methods, tenure, charges, and churn status.

Python Data Cleaning & Preprocessing

Converted TotalCharges to numeric values, handled missing values, checked duplicate records, standardized column names, created tenure groups, and exported the cleaned dataset.

SQL Analysis Performed

Calculated churn rate, analyzed churn by contract type, payment method, gender, senior citizens, internet service, revenue loss, and implemented aggregate functions, CTEs, and window functions.

Power BI Dashboard Components

Created KPI cards, donut charts, bar charts, stacked bars, line charts, treemaps, matrices, and interactive slicers for business insights.

Business Insights

Month-to-month customers tend to churn more, long-term contracts improve retention, monthly charges influence churn, and payment methods affect customer behavior.

Skills Implemented

Python, Pandas, NumPy, Data Cleaning, Feature Engineering, MySQL, SQL, CTEs, Window Functions, Power BI, DAX, Data Visualization, Business Analysis, GitHub, and Data Storytelling.

Tools & Technologies

Python, Jupyter Notebook, MySQL, Power BI, DAX, Pandas, NumPy, CSV Processing, and GitHub.

Project Outcome

The project demonstrates a complete data analytics workflow from data preparation to interactive business intelligence reporting.